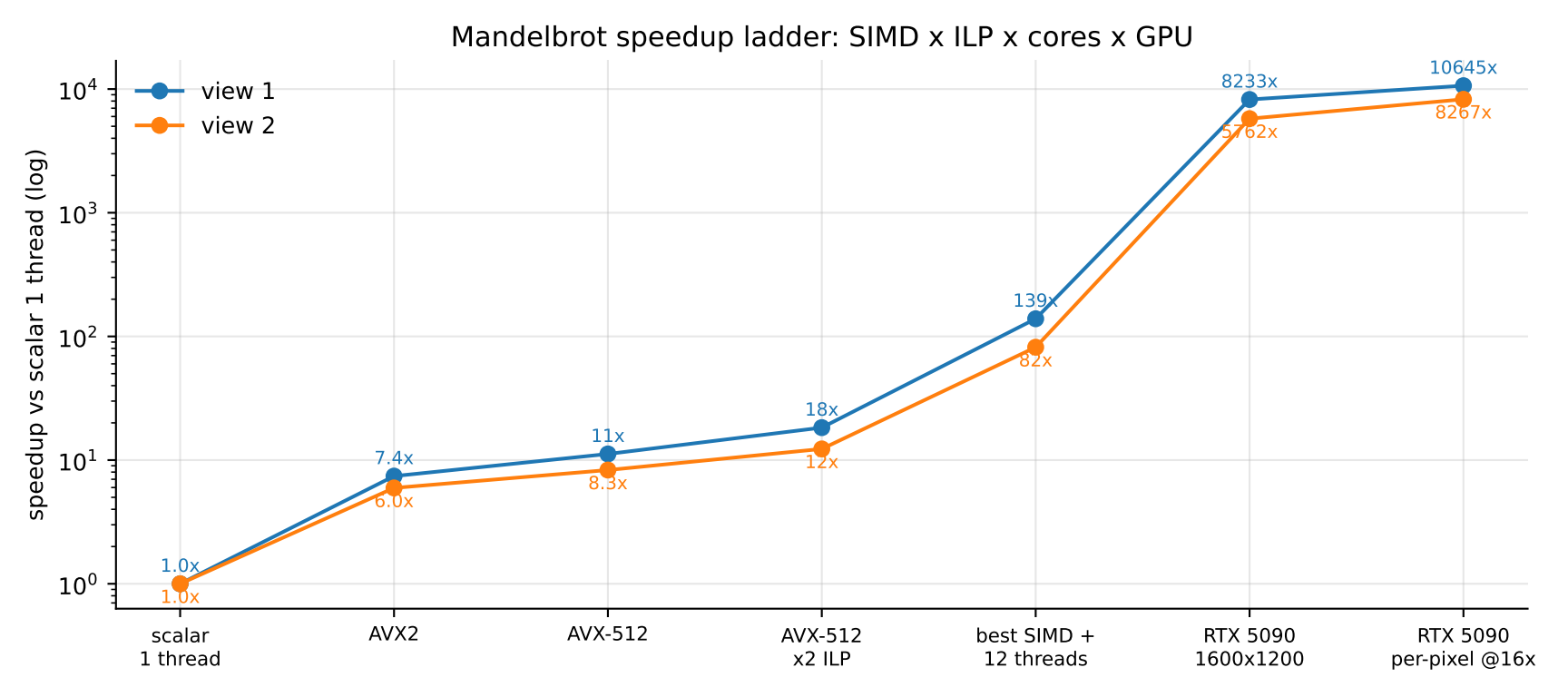
Mandelbrot speedup ladder: SIMD x ILP x cores x GPU
speedup vs scalar 1 thread (log)
view 1
view 2
10^0
10^1
10^2
10^3
10^4
1.0x
1.0x
7.4x
6.0x
11x
8.3x
18x
12x
139x
82x
8233x
5762x
10645x
8267x
scalar
1 thread
AVX2
AVX-512
AVX-512
x2 ILP
best SIMD +
12 threads
RTX 5090
1600x1200
RTX 5090
per-pixel @16x